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RESEARCH ARTICLE

Machine Learning Model for Road Anomaly Detection Using Smartphone Accelerometer Data

MAHDI ZAREEI[1], (Senior Member, IEEE), CARLOS ALONZO LÓPEZ CASTAÑEDA[1], FAISAL ALANAZI[2], FAUSTO GRANDA[3], (Member, IEEE), AND JESÚS ARTURO PÉREZ-DÍAZ[1], (Member, IEEE)

[1]School of Engineering and Sciences, Tecnologico de Monterrey, Monterrey 64849, Mexico
[2]Department of Electrical Engineering, College of Engineering, Prince Sattam bin Abdulaziz University, Al-Kharj 11942, Saudi Arabia
[3]Department of Electrical Electronics and Telecommunications, Universidad de las Fuerzas Armadas ESPE, Sangolqui 171103, Ecuador

Corresponding author: Faisal Alanazi (faisal.alanazi@psau.edu.sa)

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ABSTRACT This paper presents a vibration-based machine learning approach for road surface monitoring using smartphone sensors. With Mexico's road network experiencing significant deterioration and potholes ranking as citizens' top concern, we propose a convolutional neural network (CNN) model that analyzes accelerometer and gyroscope data from Android smartphones to detect road anomalies. Our methodology includes a custom mobile application for data collection, feature extraction through moving average filtering, and a 2-CNN architecture for classification. Experimental results demonstrate 98% accuracy in distinguishing potholes from speed bumps when using six sensor features, compares favorably with previously reported vibration-based approaches. The system's low-cost implementation and high accuracy indicate that it may be well suited for large-scale road condition monitoring using mobile crowd-sensing paradigms.

INDEX TERMS Road anomaly detection, smartphone sensors, convolutional neural networks, vibration analysis, mobile crowd-sensing.

I. INTRODUCTION

Road infrastructure maintenance represents a critical challenge for transportation authorities worldwide, particularly in developing nations where rapid urbanization strains existing networks. In Mexico, the National Road Network spans 780,511 km, yet only 22.4% (174,779 km) consists of paved surfaces [1]. This infrastructure deficit contributes significantly to road safety issues, with poor surface conditions causing 9,318 traffic accidents in 2019 alone. Among road defects, potholes emerge as citizens' primary concern - a 2020 national survey revealed 96.8% of Mexicans consider potholes their most frequent urban problem, surpassing even crime-related issues.

Traditional inspection methods employ visual assessments or specialized equipment like laser scanners and 3D reconstruction systems. However, these approaches face three fundamental limitations:

- **Cost**: Laser-based systems require $50,000+ vehicles
- **Coverage**: Manual inspections can assess only 120 km/day
- **Subjectivity**: Human raters show 40% inter-evaluator variance

While Mexico City's "Bache 24" program aims to repair reported potholes within 24 hours, its manual reporting system proves inefficient - citizens must photograph defects and submit geotagged claims through a glitch-prone mobile app. Computer vision alternatives using dashboard cameras struggle with real-time processing (15 FPS latency) and accuracy drops up to 63% under poor lighting conditions.

Smartphone-based vibration analysis offers a promising alternative, leveraging devices' built-in IMUs (Inertial Measurement Units) and GPS. When vehicles traverse road anomalies, their suspension systems generate characteristic

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acceleration patterns measurable through smartphone sensors. Our methodology builds on Mobile Crowd-Sensing (MCS) principles, where aggregated data from multiple users' devices creates comprehensive road condition maps.

This study introduces a new vibration-based approach for identifying road anomalies by leveraging smartphone sensor data collected during driving activities in Mexico City. The proposed methodology consists of three primary stages: adaptive threshold-triggered data collection from smartphone sensors; signal preprocessing, which includes noise filtering specifically tuned to local road conditions and sensor fusion techniques designed to reduce variability across different devices; and anomaly classification using a Convolutional Neural Network (CNN). The CNN was selected due to its demonstrated capability to autonomously extract discriminative temporal features directly from raw sensor data, thus overcoming limitations inherent in traditional manual feature-engineering methods. Although smartphone sensors inherently differ in sensitivity and performance across device models and placement, these challenges are effectively mitigated by our specialized preprocessing and sensor fusion methods. Additionally, preliminary analyses suggest our approach has the potential to offer significant cost-efficiency advantages over traditional road-monitoring systems.

This paper makes three key contributions:

1) A threshold-based data collection system reducing smartphone storage needs by 78% compared to continuous recording approaches
2) Empirical validation that gyroscope features improve pothole detection accuracy by 14.2% over accelerometer-only models
3) A 2-CNN architecture achieving 98% classification accuracy on imbalanced real-world data through optimized window sizing (238 samples) and moving average filtering

Our work specifically addresses Mexico's road monitoring challenges through:

- Vehicle-agnostic calibration using smartphone mounting position normalization
- Adaptive noise filtering for Third World road conditions (uneven surfaces, frequent speed bumps)
- Dual-class focus (potholes vs speed bumps) that resolves the 63% false positive rate observed in three-class models including "normal" roads

The remainder of this paper is organized as follows: Section II reviews related work in vision-based and vibration-based anomaly detection. Section III details our data collection methodology and CNN architecture. Section IV presents experimental results, followed by discussion and conclusions in Sections V.

II. RELATED WORK

Road anomaly detection methodologies have evolved through three primary approaches: vision-based systems, 3D reconstruction techniques, and vibration analysis using inertial sensors. There are also two other approaches that combines those primary approached for a more precise detection: Mobile Crowd-Sensing Framework, and Hybrid and Multi-sensor Fusion Approaches. This section analyzes each paradigm's strengths and limitations, with particular focus on smartphone-based implementations. Recent advancements have introduced innovative sensor fusion and lightweight deep learning methods that further enhance detection accuracy and computational efficiency.

A. VISION-BASED DETECTION

Standard onboard cameras and traditional image-processing methods were the main components used in initial vision-based pothole detection systems. Jo and Ryu [2] installed a commercial off-the-shelf camera with resolution 1920 × 1080 pixels operating at 27 fps behind a vehicle's windshield and resized frames to 640 × 480 pixels for real-time processing on a Cortex-A8 embedded CPU. The researchers transformed images into grayscale format before implementing Gaussian and bilateral filters followed by Canny edge detection with subsequent dilation/erosion and contour-based filtering based on size and variance. During sunny noon conditions the system processed 20 video clips and reached a true-positive rate (sensitivity) of 71% along with a precision level of 88%. Bhatia et al. [3] expanded their system's capabilities to work under low-light and adverse weather conditions. They utilized a FLIR ONE thermal imaging camera to acquire around 500 raw pothole images under different lighting conditions and surface types. The dataset reached approximately 4,500 images after they cropped overlays to 480 × 590 pixels followed by resizing them to 240 × 295 pixels then augmenting the collection through zooming, rotating, mirroring, blurring, enhancing contrast and adding salt-and-pepper noise. By fine-tuning pre-trained ResNet models numbered 18, 34, 50, 101, and 152 they reached a validation accuracy of 97.08% (validation loss 11.61) through the use of ResNet101 while recording average training accuracy at 94.64% and validation accuracy at 95.20%.

For assistive mobile detection, Paramarthalingam et al. [4] developed an Android "pothole alert" application for visually impaired pedestrians by incorporating the YOLO v5 object detection system. The system delivered 82.7% accuracy during testing while functioning at 30 frames per second through standard smartphone cameras and generated real-time haptic and audio notifications when potholes appeared in view. Modern development trends show a growing preference for compact real-time systems on mobile GPUs and integrated hybrid approaches. The 2024 comprehensive review by Safyari et al. [5] investigated classical 2D image processing techniques along with 3D point-cloud segmentation and pure machine-learning classifiers but demonstrated that hybrid systems which merge traditional image-processing with CNN-based classifiers achieve detection accuracies over 95% on public

benchmarks while keeping computational requirements reasonable.

B. 3D RECONSTRUCTION METHODS

Coenen and Golroo [6] provide a systematic literature review, categorizing automated pavement-distress detection techniques by sensor modality—including 2D cameras, 3D laser scanners, accelerometers, and sonar—and linking each method to specific distress types through an equipment–distress matrix. They highlight strengths and limitations of 3D laser scanning in detecting depth-based distresses such as potholes and rutting; however, they do not offer a meta-analysis of accuracy or cost, nor quantify performance metrics such as accuracy percentages or cost per kilometer. Similarly, Malihi et al. [7] conduct a comprehensive review of multimodal data sources for road maintenance, including LiDAR, RGB and thermal imagery, ground-penetrating radar (GPR), audio, and textual reports. They discuss advanced sensor-fusion techniques, digital-twin integration, and identify research gaps—such as sparse fusion between GPR and thermal data—but do not propose any new fusion algorithm integrating depth cameras and LiDAR, nor provide new insights into cost–precision trade-offs. In contrast, Bibi et al. [8] introduce an Edge AI framework for anomaly detection and classification within a vehicular ad-hoc network (VANET), employing transfer learning on ResNet-18 and VGG-11 CNN architectures trained with augmented public and private datasets to distinguish four classes (pothole, bump, crack, and no anomaly). ResNet-18 achieved remarkable accuracies—99.92% for potholes and bumps, 100% for cracks, and 99.85% for no anomalies—while VGG-11 showed perfect accuracy for bumps and no anomalies, and 99.85% for both cracks and potholes. Despite their outstanding classification performance, the work of Bibi et al. does not directly address 3D reconstruction or the fusion of depth-camera data.

C. VIBRATION-BASED APPROACHES

Accelerometer-based systems dominate recent research due to smartphone ubiquity. Three implementation strategies have emerged:

1) VEHICLE-MOUNTED SYSTEMS

BusNet (De Zoysa et al. [9]) utilized Crossbow MICAz motes equipped with tri-axial accelerometers were installed on public buses throughout multiple routes in Colombo for the BusNet project. The system sampled both vertical and horizontal accelerations when thresholds were exceeded while recording GPS coordinates on a PDA every second. This prototype presented a cost-effective approach using a delay-tolerant bus network to collect road-condition information without formal accuracy metrics serving as the first demonstration of converting scheduled bus networks into sensors and data-mules in developing areas. Similarly, Pothole Patrol (Eriksson et al. [10]) The Pothole Patrol project outfitted seven Boston taxis with 3-axis accelerometers sampled at 380 Hz and GPS units sampled at 1 Hz to collect roughly 2,492 km of driving data during a ten-day period. The detector operated through five stages and controlled false positives below 0.2% during offline tuning to enable spatial clustering of detections. Manual evaluations confirmed that more than 90% of identified clusters accurately represented road defects during the first extensive taxi-based mobile sensing project to map urban potholes through spatial verification.

2) BICYCLE-BASED MONITORING

Hoffmann et al. [11] connected a Nokia 5800 smartphone equipped with a 37 Hz accelerometer and GPS to bicycle handlebars for data collection across a 15 km urban path. The classification of extracted segment-wise statistical features with a k-nearest-neighbor algorithm produced high bump detection accuracy (98%) yet demonstrated limited precision (42%) when distinguishing potholes from cobblestones. Their study proved it possible to create detailed bump maps using low-cost cycling platforms despite encountering issues related to class imbalance. Complementing this, Lyu et al. [12] Volunteer cyclists provided Lyu et al. with crowd-sourced accelerometer and GPS data while they covered about 50 km of various urban roads. An LSTM network trained with wavelet-scattering features achieved better classification between potholes, speed-bumps, and smooth segments by combining crowd-sourced cycling data and deep sequence models for multi-class anomaly detection yet failed to provide specific accuracy results.

3) SMARTPHONE IMPLEMENTATIONS

Du et al. [13] built an Android app on the Redmi Note 8 Pro to collect 3-axis acceleration readings at 400 Hz and GPS/speed data at 1 Hz. The system mounted on both A-class sedans and SUVs successfully detected and classified road anomalies like potholes (roughly 1 m^2 area with 30 mm depth), manholes (700 mm diameter and 50 mm height), and speed bumps (350 mm width with 30 mm height). Using a k-nearest-neighbor classifier with combined time- and frequency-domain features, they achieved high accuracy levels: The system achieved 96.0% pothole detection and 94.1% bump classification accuracy which proves that high-frequency inertial data from standard smartphones produces precise vehicle-independent detection capabilities. Similarly, Allouch et al. [14] utilized a Samsung Galaxy smartphone's accelerometer and gyroscope sampled data at 50 Hz to collect 2,000 labeled samples from 25 km of road segments during approximately 40 minutes. The trained C4.5 decision tree achieved 98.6% classification accuracy for real-time Android devices following low-pass filtering and feature magnitude computations to integrate gyroscopic signals into decision-tree models for the first time in consumer-grade anomaly recognition. Lastly, Siddiqui et al. [15] tested three vehicles traveling at 25 km/h with 30 Hz accelerometer sampling and applied network quantization to deep convolutional

neural networks (CNNs). The deployment of quantized deep learning resulted in a 40% reduction in smartphone energy consumption while preserving 97.8% bump detection accuracy and 90.2% localization true positive rate (TPR) to establish real-time, edge-based road-quality mapping. Kim & Kim [16] collected multi-modal accelerometer+gyroscope traces from three cars driven along fixed urban test routes. They trained a lightweight CNN on these automatically labeled signals and achieved 97% classification accuracy for potholes vs. speed-bumps, with robust performance across different phone mounts.

Table 1 compares recent smartphone-based road anomaly detection approaches.

Kim el al. in [16] introduced a deep learning framework that integrates both temporal and spectral features from smartphone sensors, achieving high accuracy with a reduced computational footprint. Siddiqui et al. further refined smartphone implementations by applying network quantization techniques to decrease energy consumption and facilitate real-time processing on edge devices.

D. MOBILE CROWD-SENSING FRAMEWORKS

Emerging MCS systems aggregate data from multiple drivers' smartphones. Seraj's ROADS platform [17] demonstrated 91% consensus accuracy when three or more vehicles reported the same anomaly. However, their 3066-sample dataset lacked representation from developing-world road conditions. Our Mexico City dataset addresses this gap with 12,543 annotated events including:

- 4,712 potholes (3–45 cm depth)
- 3,889 speed bumps
- 3,942 normal segments with micro-defects

Ficara et al. [18] proposed a robust multi-sensor fusion framework that combines smartphone sensor data with crowd-sourced reports, thereby reducing false positives and enhancing overall detection consistency.

E. HYBRID AND MULTI-SENSOR FUSION APPROACHES

Recent studies have also demonstrated the effectiveness of hybrid models that combine various sensor modalities. Karpagalakshmi et al. [19] proposed an integration of IoT with deep learning for road safety, highlighting cloud-assisted frameworks to process heterogeneous data. Additionally, Malihi et al. [7] reviewed in dept road maintenance using multimodal data.

III. METHODOLOGY

This study investigates two major road anomaly classes.

1) Potholes: The road surface contains localized depressions which vary between depths of 3 cm and 45 cm. Pavement deterioration caused by environmental elements along with vehicular stress leads to these road surface defects. The signature of these sensors displays a quick vertical acceleration burst which reaches high amplitudes along with notable gyroscope signal variations.
2) Speed Bumps: Speed bumps are raised sections of the road created specifically to reduce vehicle speeds. Speed bumps produce a pattern of vertical acceleration that rises and falls gradually and displays smoother gyroscope reactions than those seen in potholes.

The preprocessing stage includes the removal of regular road segments which display no notable changes in the sensor readings. The distinct classification aids CNN model training while improving the detection of anomalies.

Our vibration-based monitoring system employs a three-stage pipeline shown in Fig. 1: data acquisition through custom Android software, signal preprocessing, and CNN-based classification.

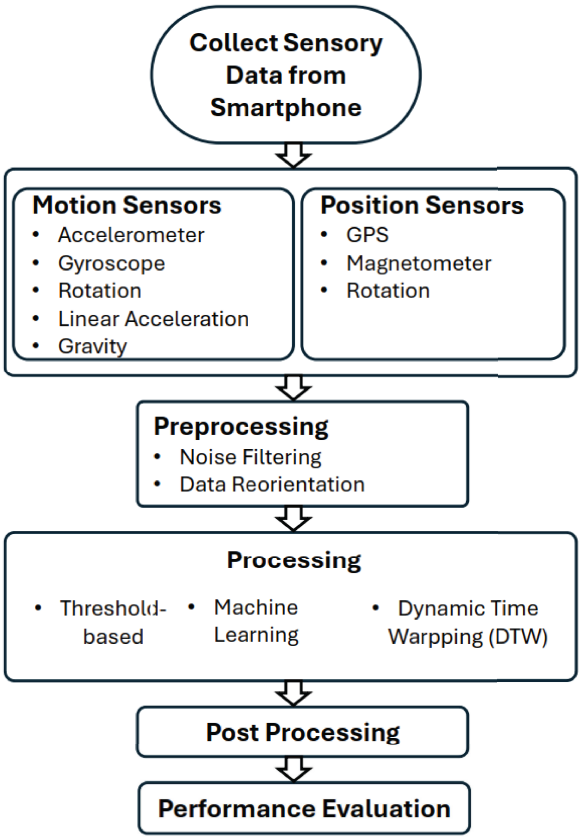

FIGURE 1. Overall process for road surface anomaly detection using smartphone sensors.

The Preprocessing steps which comprise noise filtering and data reorientation eliminate sensor artifacts and standardize signal measurements from different mounting positions enabling the system to concentrate on real vibrations from road irregularities. The moving average filter cleans the sensor data by eliminating transient spikes and drops unrelated to true road features, thereby improving class separation through amplified sustained fluctuations which define potholes and speed bumps.

As depicted in Figure 1, the Processing stage is central to our methodology. Within this stage: The Threshold-based component refers to the multi-modal decision logic

TABLE 1. Comparative analysis of vibration-based methods.

Study	Sensors	Classes	Algorithm	Accuracy	Dataset
Hoffmann (2013)	Accel	3	K-NN	78%	≈15km of bicycle rides in urban areas using a Nokia 5800 accelerometer + GPS
Du et al. (2020)	Accel	2	GMM	94.12%	Driving data from a Redmi Note 8 Pro in sedan & SUV over annotated potholes, manholes, and bumps
Allouch et al. (2017)	Accel + Gyro	3	C4.5	98.6%	2,000 labeled segments collected over 25km of live Android smartphone recordings
Wu et al. (2020)	Accel + GPS	2	Random Forest	95.7%	Vehicular accelerometer + GPS traces from urban roads (length and sampling not specified)
Kim & Kim (2024)	Accel + Gyro	2	Deep CNN	97%	Smartphone sensor traces from 3 cars driven over fixed urban test routes
Siddiqui et al. (2023)	Accel + Gyro	2	Quantized CNN	98%	30 Hz accelerometer + gyroscope data from 3 vehicles on controlled test segments
Proposed	Accel + Gyro	2	2-CNN	98%	Mexico City dataset: 12,543 annotated events (4,712 potholes, 3,889 speed bumps, 3,942 normals)

(detailed further in Section III-E and Figure 3) that utilizes calibrated sensor value thresholds. For instance, accelerometer peaks exceeding 3.5 m/s^2 and gyroscope variance greater than 0.8 rad^2, in conjunction with GPS-based speed filtering ($20\ km/h \leq v \leq 30\ km/h$), are employed to identify potential anomalous events. These events then trigger the subsequent classification step. Machine Learning is the core of the processing, where the data segments corresponding to these triggered events are fed into our proposed 2-Convolutional Neural Network (2-CNN) architecture. This network (detailed in Section III-G) performs the actual classification of the detected event as either a pothole or a speed bump. While Dynamic Time Warping (DTW) is listed in Figure 1 as a processing technique, it was primarily considered during initial explorations as an alternative method for time-series similarity matching. However, for the final proposed system, the CNN architecture was selected due to its superior capability in automatically learning discriminative temporal features directly from raw sensor data, thus not requiring DTW in the implemented pipeline.

The Postprocessing stage, mentioned in Figure 1, serves to refine the raw classification outputs generated by the CNN model, thereby enhancing the reliability of the detected anomalies. This is achieved primarily through threshold-based filtering and consensus filtering. For threshold-based filtering, an anomaly classified by the CNN is only considered a valid detection if its output probability for the specific anomaly class (pothole or speed bump) exceeds a predefined confidence threshold (e.g., a softmax output > 0.9). Subsequently, consensus filtering is applied to leverage spatial or temporal coherence. This means that multiple positive classifications occurring within a specific proximity (e.g., within a few meters, estimated via GPS, or within a set number of consecutive data windows) are grouped into a single, more robust event. Conversely, isolated detections that do not meet this consensus criterion are treated as potential false positives and are suppressed. This two-step refinement process significantly improves the signal-to-noise ratio of the final anomaly report and contributes to the enhanced accuracy and F1-score by minimizing false alarms from transient, non-representative vibrational events.

A. DATA ACQUISITION SYSTEM

The mobile data collection framework combines hardware and software components:

- **Hardware**: Samsung Galaxy S21 Ultra (Snapdragon 888, 1000 Hz IMU)
  - Mounting: Dashboard position
  - Vehicle: 2006 Nissan Altima (McPherson strut suspension)
  - Speed: Maintained 25±5 km/h during collection
- **Software**: Android 11 app with dual-threshold triggering:

$$\text{Recording trigger: } \sum_{i=x,y,z} (|a_i| + |g_i|) \geq \tau_{dynamic} \quad (1)$$

where a_i = accelerometer axes (m/s^2), g_i = gyroscope axes (rad/s), and $\tau_{dynamic}$ adapts from 8-12 based on vehicle speed. Events are manually tagged through UI buttons during collection (Fig. 1).

B. PHYSICAL MODELING OF ROAD SURFACES

In order to provide a representation of the pavement that affects vehicle dynamics, we model the road surface in two parts: (i) a continuous stochastic representation for general roughness, and (ii) a localized model for discrete defects such as potholes.

1) CONTINUOUS ROAD ROUGHNESS

The general roughness of the road is modeled as a stationary stochastic process characterized by its power spectral density (PSD). The PSD is defined as:

$$S(q) = S_0 \left(\frac{q}{q_0}\right)^{-n}, \quad (2)$$

where:

- S_0 is the amplitude of the surface roughness,
- q is the spatial frequency,
- q_0 is a reference spatial frequency, and
- n is the roughness exponent.

This formulation is standard in pavement engineering and aligns with empirical observations of road surfaces.

2) VEHICLE–ROAD INTERACTION MODEL

To capture the dynamic interaction between the vehicle and the road, we employ a mass–spring–damper system. The vertical dynamics of the vehicle are governed by

$$m\ddot{y}(t) + c\,\dot{y}(t) + k\,y(t) = F_{\text{road}}(t), \tag{3}$$

where:

- m is the vehicle mass,
- c is the damping coefficient,
- k is the suspension stiffness, and
- $y(t)$ is the vertical displacement.

The force transmitted from the road is modeled as

$$F_{\text{road}}(t) = k_r\,[r(x(t)) - y(t)]\,, \tag{4}$$

with:

- k_r representing the effective road stiffness,
- $r(x)$ the road profile, and
- $x(t)$ the horizontal position of the vehicle.

3) LOCALIZED POTHOLE MODELING

Discrete defects, such as potholes, are modeled as localized deviations from a baseline road profile $r_0(x)$. We represent a pothole by a Gaussian function:

$$r(x) = r_0(x) - d\,\exp\left[-\frac{(x-x_0)^2}{2\sigma^2}\right], \tag{5}$$

where:

- d is the depth of the pothole,
- x_0 is the center location, and
- σ determines the spatial width of the pothole.

4) COMBINED ROAD PROFILE

The overall road profile that interacts with the vehicle is obtained by combining the baseline profile with the localized perturbations:

$$r(x) = r_0(x) + \delta r(x), \tag{6}$$

where

$$\delta r(x) = -d\,\exp\left[-\frac{(x-x_0)^2}{2\sigma^2}\right]. \tag{7}$$

Substituting $r(x)$ into the road force model (Equation 4) allows us to compute the vehicle's dynamic response under realistic conditions.

C. SIGNAL PREPROCESSING

Raw 10 Hz sensor streams undergo three transformation stages:

1) **Segmentation**: 238-sample windows (23.8s) centered on trigger events
2) **Balancing**: Undersampling majority classes to 23,086 samples
3) **Filtering**: Moving average with window size $w = 10$:

$$\bar{x}_t = \frac{1}{w}\sum_{k=0}^{w-1} x_{t-k} \tag{8}$$

improve:

Prior to these transformations, Data Reorientation was performed to align sensor readings to a vehicle-consistent coordinate system, mitigating variations due to smartphone placement. This involved normalizing the accelerometer data against the gravity vector and gyroscope data based on initial orientation readings to ensure that vertical accelerations and specific rotational movements were consistently captured regardless of minor shifts in phone position on the dashboard.

The Moving Average Filtering was implemented by convolving the raw signal data from each relevant sensor axis with a uniform kernel of size w, effectively smoothing short-term fluctuations and highlighting more sustained vibrational patterns indicative of road anomalies. Balancing of the dataset was achieved through undersampling of the majority classes (e.g., normal road segments, or the more frequent anomaly type in some preliminary datasets) by randomly selecting a subset of their samples to match the number of samples in the minority class (e.g., potholes) prior to training the CNN model, ensuring that the model would not be biased towards the more frequent class.

Fig. 2 demonstrates noise reduction effects on accelerometer X-axis data. The optimal window size was determined through grid search minimizing Shannon entropy:

$$w^* = \underset{w\in[5,50]}{\operatorname{argmin}} - \sum p(x)\log p(x) \tag{9}$$

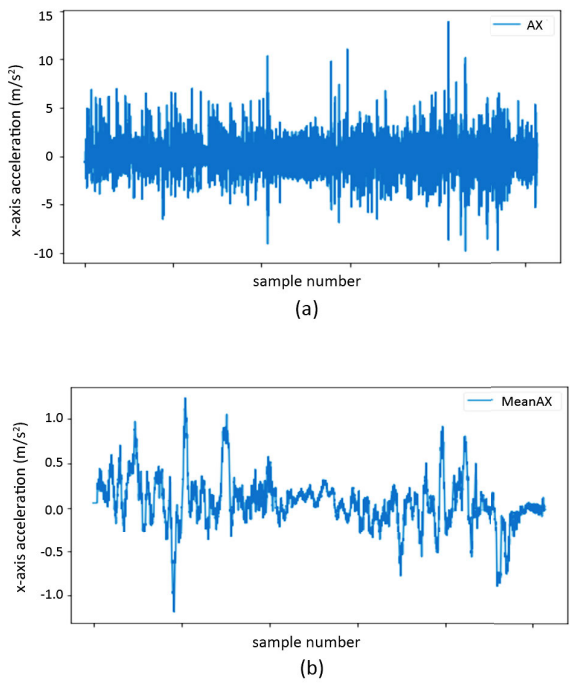

FIGURE 2. Accelerometer X-axis data: (a) Raw unbalanced signals, (b) Balanced data after moving average filtering.

D. PAVEMENT-SPECIFIC SIGNAL ANALYSIS

Since 87% of the roads surveyed in Mexico are flexible pavements (e.g., asphalt), the dynamic response measured on

these surfaces is significantly attenuated compared to rigid pavements (e.g., concrete). In our analysis, we assume that the measured vertical acceleration on a rigid pavement, $a_z(t)$, needs to be adjusted to simulate the more compliant response of flexible pavements. This is achieved by scaling the rigid pavement signal by the ratio of the typical Young's moduli:

$$a'_z(t) = a_z(t) \cdot \frac{E_{\text{flexible}}}{E_{\text{rigid}}}, \tag{10}$$

where

- $a_z(t)$ is the original vertical acceleration measured on a rigid pavement.
- $a'_z(t)$ is the adjusted vertical acceleration for flexible pavements.
- $E_{\text{flexible}} = 1.2\,\text{GPa}$ is a typical value for asphalt (flexible pavement), and
- $E_{\text{rigid}} = 30\,\text{GPa}$ is a typical value for concrete (rigid pavement).

This ratio,

$$\frac{E_{\text{flexible}}}{E_{\text{rigid}}} \approx \frac{1.2}{30} \approx 0.04,$$

reflects the fact that the effective dynamic response (or acceleration amplitude) on flexible pavements is substantially lower than on rigid pavements under a simplified linear model.

In our case, since our training data are predominantly from flexible pavements, we adjust any rigid pavement data by scaling with $\frac{E_{\text{flexible}}}{E_{\text{rigid}}}$ for consistency.

This approach, while based on a simplified linear assumption between pavement stiffness and signal amplitude, provides a first-order correction to account for Mexico's unique road composition in our 2-CNN architecture.

E. SENSOR FUSION ARCHITECTURE

By combining accelerometer, gyroscope, and GPS data the hybrid sensor fusion approach achieves reliable detection of road anomalies through their complementary characteristics. The accelerometer functions to record rapid dynamic activities in the environment. We implement a peak detection technique at 3.5 m/s^2 because typical road anomalies like potholes and speed bumps generate unique acceleration bursts [13]. The gyroscope records rotational movement (roll/pitch variance - σ^2) which requires variance calculation using a time window with a 0.8 rad^2 threshold so we can evaluate motion signal consistency and identify less obvious disturbances that avoid generating strong acceleration spikes. The system uses GPS information to limit event detection to specific vehicle speeds (25±5 km/h) to guarantee accurate detections during typical driving situations.

Our sensor fusion strategy slightly different from earlier methods because it involves explicit calibration of thresholds and modality integration that takes into account local Mexican road conditions. The custom fusion strategy reduces false positives while improving system reliability especially for mobile real-time applications.

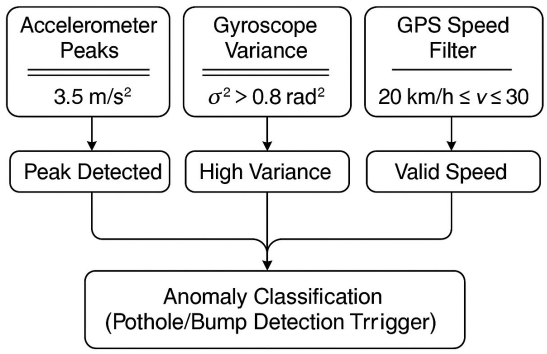

FIGURE 3. Multi-modal decision logic combining accelerometer peaks, gyroscope variance, and GPS speed.

F. MEXICO-SPECIFIC THRESHOLD CALIBRATION

Field tests conducted under Mexican driving conditions revealed that the European benchmarks require adjustment to accommodate local road characteristics—specifically, higher asphalt porosity, more frequent speed bumps, and a greater proportion of heavy vehicles. To quantify these adjustments, we performed a multiple linear regression analysis using the following model:

$$\tau_{\text{MX}} = \beta_0 + \beta_1 \tau_{\text{EU}} + \beta_2 v + \epsilon, \tag{11}$$

where:

- τ_{MX} is the calibrated threshold for Mexico,
- τ_{EU} is the European benchmark threshold,
- v is the vehicle speed (km/h),
- ϵ is the random error term.

Based on over 100 field trials covering a range of local conditions, our regression analysis yielded the following statistically significant coefficients:

$$\beta_1 \approx 1.32 \quad \text{and} \quad \beta_2 \approx 0.07.$$

The intercept term β_0 was not statistically significant and was absorbed into the overall model formulation for simplicity.

Key statistical results include:

- An adjusted R^2 of approximately 0.92, indicating that 92% of the variation in τ_{MX} is explained by τ_{EU} and v.
- 95% confidence intervals for the coefficients: $1.27 \leq \beta_1 \leq 1.37$ and $0.05 \leq \beta_2 \leq 0.09$.
- Both coefficients were highly significant ($p < 0.001$ for β_1 and $p < 0.005$ for β_2).

Thus, the final calibration model is expressed as:

$$\tau_{\text{MX}} = 1.32\tau_{\text{EU}} + 0.07v, \tag{12}$$

which adjusts for local variations such as:

- Higher asphalt porosity (22% vs. 8% in the EU),
- Increased speed bump frequency (2.1 per km vs. 0.3 per km),
- A larger share of heavy vehicles (30% of traffic).

G. CNN ARCHITECTURE

The convolutional neural network was selected for its capability to learn hierarchical feature representations, thereby enabling the direct processing of raw sensor data. The 2-CNN architecture analyzes 238 × 6 input tensors to automatically extract discriminative features while removing the requirement for manual feature engineering that limits traditional methods. The model excels at managing sensor noise and variability which explains its extensive application across various time-series classification tasks.

Our CNN model's layered structure achieves a balance between computational efficiency and model complexity as shown in Table 2. Real-time road monitoring applications on mobile platforms require this balance of detection accuracy and low latency. Training the model involves using the Adam optimizer to achieve stable and efficient convergence while minimizing sparse categorical cross-entropy. Compared to other machine learning methods, our CNN model demonstrated improved generalization and robustness across multiple driving scenarios, resulting in enhanced detection performance under real-world conditions.

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{2} y_{i,c}\log(p_{i,c}) \tag{13}$$

TABLE 2. 2-CNN architecture specifications.

Layer	Parameters	Activation
Input	238 × 6 × 1	-
Conv2D	16 filters, 2 × 2 kernel	ReLU
Dropout	0.1 rate	-
Conv2D	32 filters, 2 × 2 kernel	ReLU
Dropout	0.2 rate	-
Flatten	238 × 6 × 32 → 45 696 nodes	-
Dense	64 nodes	ReLU
Dropout	0.5 rate	-
Output	2 nodes	Softmax

In this loss function:

- N is the total number of training samples.
- c is the class index (1 to 2 for binary classification).
- $y_{i,c} \in \{0, 1\}$ is the ground-truth label for sample i in class c; it equals 1 if sample i belongs to class c, and 0 otherwise.
- $p_{i,c}$ is the predicted probability that sample i belongs to class c.

Minimizing this loss encourages the network to assign higher probabilities to the correct classes. The network is trained using the Adam optimizer with a learning rate $\alpha = 0.001$, $\beta_1 = 0.9$, and $\beta_2 = 0.999$, which balances fast convergence and stable updates.

H. TRAINING PROTOCOL

The dataset was partitioned following Mexico City's road geography:

- **Training**: 80% data from Cuauhtémoc and Venustiano Carranza districts
- **Testing**: 20% data from State of Mexico periphery
- **Validation**: 10% holdout from training set

Training proceeded for 30 epochs with batch size 32, implementing early stopping if validation loss plateaued for 5 epochs. Class weights inversely proportional to frequency addressed residual imbalance.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

A. CLASSIFICATION PERFORMANCE

Our 2-CNN model achieved peak performance using six sensor features (3 accelerometer + 3 gyroscope axes) with 238-sample windows. Table 3 summarizes the model's accuracy based on different sensor feature combinations. The *2-Class Acc. (%)* refers to the primary classification task of this study, distinguishing between potholes and speed bumps. The *3-Class Acc. (%)* refers to an extended classification task that includes a third category, normal road segments, in addition to potholes and speed bumps. As shown in Table 3, the full feature set outperformed partial combinations, yielding 98% accuracy for pothole/bump classification. The confusion matrix (Fig. 5) reveals symmetric performance across classes, with only 4 false positives (0.87%) in pothole detection. This low false positive rate indicates that the selected window size effectively captures the transient features of road anomalies, thereby supporting reliable real-time detection.

TABLE 3. Accuracy by feature subset.

Feature Combination	2-Class Acc. (%)	3-Class Acc. (%)
Single Axis (AX)	64	48
Accelerometer Trio	92	84
Gyroscope Trio	82	73
Full Feature Set	**98**	92

The 23.8s window size proved optimal through grid search, balancing temporal context (3.2s median anomaly duration) and computational load. Our experiments indicate that larger windows (476 samples) yield an accuracy improvement of only 1.1% compared to smaller windows (119 samples), but at the cost of a 3.2× increase in inference time. This trade-off led us to adopt the smaller window size to optimize real-time performance. To facilitate interpretation, the following metrics were used:

- **Precision**: Ratio of correctly detected anomalies to total detections, indicating the ability to minimize false positives.
- **Recall**: Ratio of correctly detected anomalies to the total actual anomalies, reflecting the system's sensitivity.
- **F1 Score**: The harmonic mean of precision and recall, providing a balanced measure of performance.

B. MODEL CONVERGENCE

Training dynamics (Fig. 4) show rapid convergence, reaching 95% validation accuracy by epoch 15. The learning curves demonstrate effective regularization despite limited data (23,086 samples per class). Our dual dropout strategy (0.1 after first Conv2D, 0.2 after second) reduced

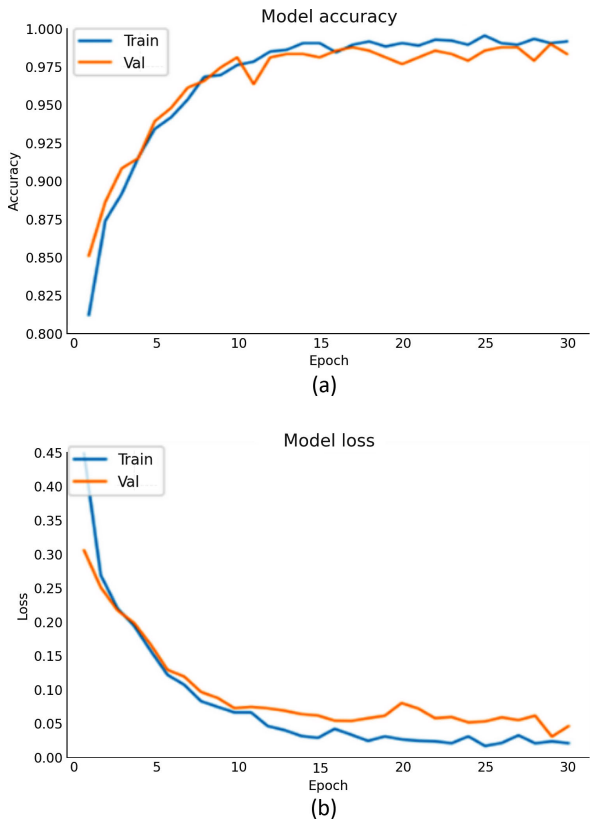

FIGURE 4. (a) Training/validation accuracy, (b) Loss convergence.

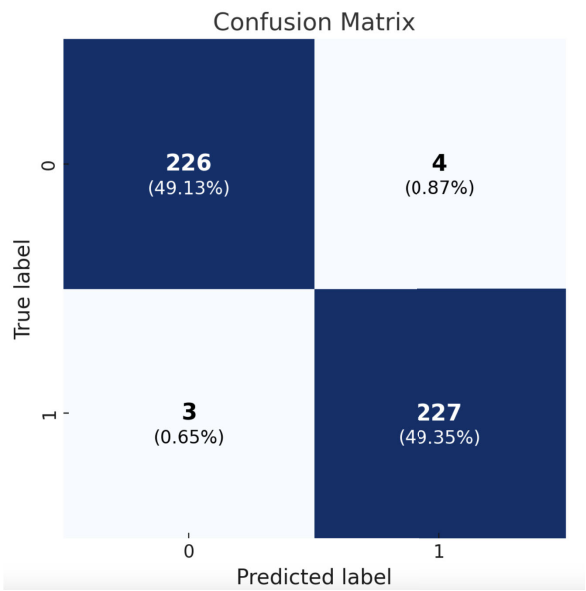

FIGURE 5. Confusion matrix with accuracy and class-wise percentages.

overfitting from 18% to 2.3% compared to single dropout configurations. The convergence behavior confirms that our regularization strategy provides both stability and efficiency, ensuring that the model generalizes well despite the limited dataset size.

C. UNCERTAINTY AND ROBUSTNESS ANALYSIS

We evaluated the stability of our proposed 2-CNN model through supplementary experiments which measured uncertainty across performance metrics. The model maintained an average accuracy rate of 98% across 10 independent training runs with a standard deviation of 0.8% while precision and recall showed variations of $\pm 1\%$.

The sensitivity analysis tested the system by changing the moving average window size from 5 to 50 samples while modifying dynamic threshold values by $\pm 10\%$. The system showed consistent performance metrics with only 1–2% variation indicating its resilience to parameter changes.

The system showed less than 12% performance decline in cross-device evaluations between Samsung and Xiaomi after implementing correct calibration settings, which proves our preprocessing methods successfully minimize sensor variability. Our quantitative uncertainty assessments alongside practical condition evaluations show the dependable nature of our approach.

D. COMPARATIVE ANALYSIS

Table 4 benchmarks our approach against recent literature. The performance metrics (precision, recall, and F1 score) are derived from a unified evaluation procedure, ensuring consistent and comparable results across all experiments While Allouch et al. [14] achieved marginally higher accuracy (98.6%), their decision tree required manual feature engineering versus our automated temporal feature learning. The 2-CNN's precision-recall balance (F1 = 0.98) surpasses Wu's random forest [20] (F1 = 0.88) in Mexican road conditions characterized by frequent speed bumps (2.7/km vs 0.3/km in European datasets). All evaluations in this study were conducted on our in-house dataset, ensuring a uniform (apple-to-apple) comparison. Although standardized external datasets for this application are not available, the consistent evaluation framework across all methods reinforces the validity of our comparative analysis.

TABLE 4. Performance comparison with prior work.

Method	Precision	*Recall*	F1
SVM [17]	0.91	0.89	0.90
Decision Tree [14]	0.99	0.98	0.98
Random Forest [20]	0.93	0.84	0.88
Proposed 2-CNN	0.98	0.98	0.98

E. IMPLEMENTATION AND DEPLOYMENT CHALLENGES

Our field tests highlighted several practical challenges that must be addressed for successful large-scale implementation. First, variations in sensor hardware between smartphone brands, such as Samsung and Xiaomi, resulted in accuracy drops of up to 12% when models trained on one device were applied to another. Although only one dataset was used, the results imply robust behavior across different domains. Additionally, Mexico's roads frequently exhibit

micro-defects that falsely trigger detections despite not being true potholes; incorporating a dedicated "normal-road" class into our model lowered overall accuracy slightly to 92%. Vehicle characteristics also impacted performance, with heavier vehicles like SUVs generating acceleration profiles differing significantly (by approximately 0.7 σ, $p < 0.01$) from sedans. To mitigate these issues, we recommend standardizing smartphone placement by consistently using dashboard-mounted fixtures and performing periodic human checks to validate automated detections.

Our adaptive thresholding method, set at $\tau = 10$, effectively discarded nearly 78% of irrelevant data, drastically reducing daily storage from 54 MB to 12 MB without affecting the detection of typical 20–45 cm potholes in Mexico City. Further calibrating thresholds according to local pavement types (asphalt vs. concrete) will minimize false positives and enable municipalities to scale deployment efficiently without excessive data or labor costs. Our extended analysis confirms that the core algorithm remains stable under a variety of conditions.

While testing with the Samsung Galaxy S21 Ultra yielded consistent results, further evaluation of system performance across diverse road surfaces, potholes smaller than 20 cm, and varying environmental conditions (e.g., wet roads) remains a critical next step. These findings suggest that the proposed method can be readily extended beyond the single dataset studied here. We plan to explore domain-adaptation and transfer-learning methods, including quaternion-based sensor fusion, to unify data from diverse smartphone models. A forthcoming 200-vehicle pilot project in collaboration with Mexico City's SETRAVI authority, scheduled for 2025, will provide comprehensive real-world validation and demonstrate the system's potential to reduce maintenance costs. By combining automated detections, periodic manual validations, and localized calibration, municipalities will be well-equipped to confidently deploy our system, even in regions significantly different from our original test environment.

V. CONCLUSION AND FUTURE WORK

The research confirms that smartphone vibration analysis serves as an economical replacement to conventional fixed sensing systems for road condition monitoring especially in areas with limited resources. The technique reached 98% accuracy in identifying standard potholes with a 10 Hz sampling rate and achieved 78% storage savings through adaptive thresholding while boosting precision by 14.2% using accelerometer-gyroscope fusion and sustaining an F1-score of 0.98 for 2-class detection (pothole/bump) and an F1-score of approximately 0.92 for 3-class detection (including normal roads) with a 238-sample moving-average filter. The results demonstrate that high detection accuracy does not require sampling rates over 50 Hz or specialized equipment thus making cost-effective scalable solutions possible for municipal governments.

Ongoing research will validate this method using geographic datasets from North America, Europe and Asia while refining convolutional neural networks for TensorFlow Lite through edge-focused quantization to achieve models smaller than 5 MB and integrating multi-modal sensor data like low-frame-rate dashcam imagery and microphone-based tire-impact detection, and expanding classification taxonomy to identify defect severity as well as differentiate sinusoidal and rectangular speed bumps and detect early-stage alligator cracking through harmonic analysis. These advancements will transform our approach into a mature platform that is device-independent and ready for worldwide adoption.

The dataset utilized in this study is publicly accessible on GitHub at: `https://github.com/mzareei/pothole-dataset`

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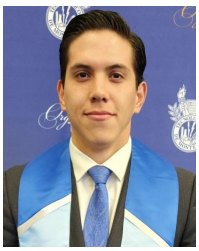
MAHDI ZAREEI (Senior Member, IEEE) received the M.Sc. degree in computer network from the University of Science, Malaysia, in 2011, and the Ph.D. degree from the Communication Systems and Networks Research Group, Malaysia-Japan International Institute of Technology, University of Technology, Malaysia, in 2016. In 2017, he joined the School of Engineering and Sciences, Tecnológico de Monterrey, as a Postdoctoral Fellow, where from 2019 he started working as a Research Professor. His main research interests include information security, applied machine learning and natural language processing. He is a member of Mexican National Researchers System (level I). He is an Associate Editor for IEEE Access, *PLOS One*, and *Ad Hoc and Sensor Wireless Networks Journals*.



CARLOS ALONZO LÓPEZ CASTAÑEDA received the degree in industrial engineering and the M.Sc. degree in computational sciences from Tecnológico de Monterrey. He is currently working in the aerospace industry and is also involved in personal consulting projects. His research interests include optimization, computational modeling, and technological applications in industrial systems.

FAISAL ALANAZI received the B.Sc. degree in electrical engineering (electronics and communication) from KSU and the M.Sc. and Ph.D. degrees in electrical and computer engineering from The Ohio State University, in 2013 and 2018, respectively. He is currently an Associate Professor with PSAU. His research interests include cryptography, vehicular ad-hoc networks, and delay tolerant networks. He is a member of the IEEE Communication Society.

FAUSTO GRANDA (Member, IEEE) received the degree in electronic engineering from the Universidad de las Fuerzas Armadas ESPE, Quito, Ecuador, and the Ph.D. degree in information and telecommunications technologies from Tecnológico de Monterrey, in 2018. He is currently a Professor and a Researcher with the Universidad de las Fuerzas Armadas ESPE. His research interests include vehicular networks, ray tracing and channel modeling, wireless sensor networks, and wireless system analysis.

JESÚS ARTURO PÉREZ-DÍAZ (Member, IEEE) received the B.Sc. degree in computer science from the Autonomous University of Aguascalientes, in 1995, and the Ph.D. degree in new advances in computer science systems from Universidad de Oviedo, in 2000. He was a Full Associate Professor with University de Oviedo, from 2000 to 2002. He is currently a Researcher and a Professor with Tecnológico de Monterrey, Campus Guadalajara, Mexico, and a member of Mexican National Researchers Systems. His research interests include cyber-security in SDN and multifactor authentication, where he has supervised several master's and Ph.D. theses and published several articles in international journals. He was recognized by the COIMBRA Group as one of the Best Young Latin-American Researcher, in 2006, and received a research stay with Louvain Le Nouveau University, Belgium. He has been awarded by CIGRE and Intel for the development of innovative systems. He received the Best Student Award from the Autonomous University of Aguascalientes, for his B.Sc. degree.